

GAN Galaxy Generation Project

Introduction

Driving Question: How Effectively can a General Adversarial Network (GAN) replicate galaxy morphology?

Process:

- Development of a GAN model (Discriminator + Generator) trained on SDSS galaxy data
- Analysis of GAN results using Sersic light profile fitting and radial color analysis of GAN results vs training sample to measure GAN Success
- Possibly in the future: Development of a CNN to further analyze GAN images and study GAN's success in matching morphology of training sample

GAN Model Architecture

Generator:

- Standard Generator Model based off of torch tutorial: 5 Conv2D layers using nn.sequential architecture
- 9.48M parameters: quite a deep/complex model
- Output: 64x64x3 color image
- .0002 Learning Rate

```
class Generator(nn.Module):
    def __init__(self, nz, ngf, nc):
        super(Generator, self).__init__()
        self.main = nn.Sequential(
            nn.utils.spectral_norm(nn.ConvTranspose2d(nz, ngf * 8, 4, 1, 0, bias=False)), # 4x4
            nn.BatchNorm2d(ngf * 8),
            nn.ReLU(True),

            GenResBlock(ngf * 8),

            nn.utils.spectral_norm(nn.ConvTranspose2d(ngf * 8, ngf * 4, 4, 2, 1, bias=False)), # 8x8
            nn.BatchNorm2d(ngf * 4),
            nn.ReLU(True),

            GenResBlock(ngf * 4),

            nn.utils.spectral_norm(nn.ConvTranspose2d(ngf * 4, ngf * 2, 4, 2, 1, bias=False)), # 16x16
            nn.BatchNorm2d(ngf * 2),
            nn.ReLU(True),

            nn.utils.spectral_norm(nn.ConvTranspose2d(ngf * 2, ngf, 4, 2, 1, bias=False)), # 32x32
            nn.BatchNorm2d(ngf),
            nn.ReLU(True),

            nn.utils.spectral_norm(nn.ConvTranspose2d(ngf, nc, 4, 2, 1, bias=False)), # 64x64
            nn.Tanh()
        )

    def forward(self, input):
        return self.main(input)
```

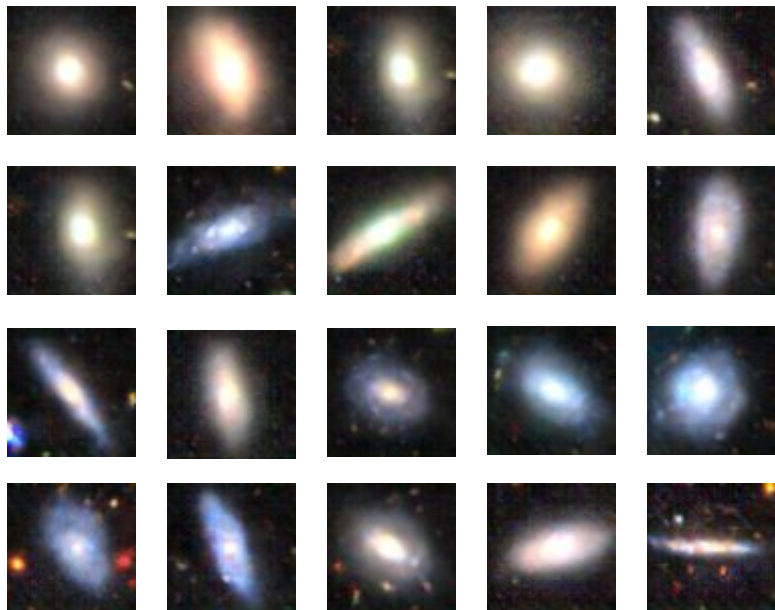
Discriminator:

- Similar to Generator Model, 5 Conv2D layers using a sequential model
- 2.7M parameters, also a deep model
- Input: 128x128x3 training image, outputs a feature array with 512 features
- .00075 Learning Rate: smaller than generator to prevent overtraining

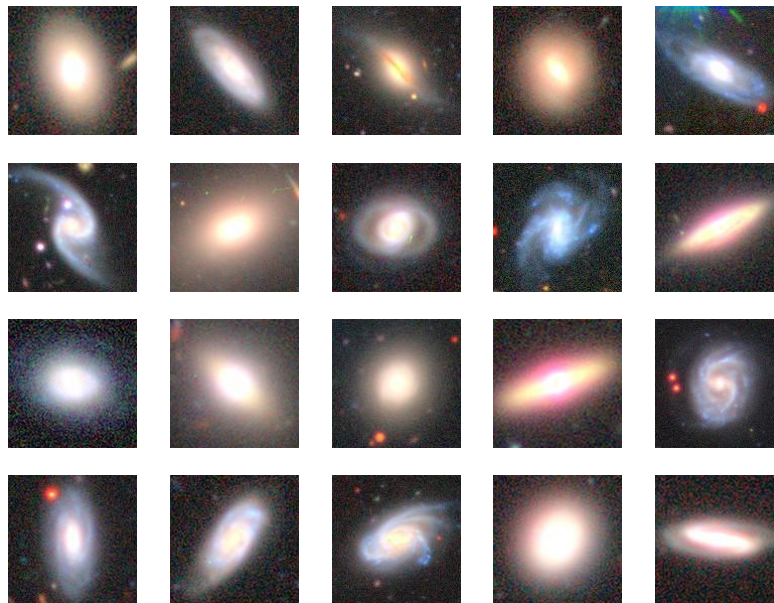
```
Discriminator(
  (main): Sequential(
    (0): Conv2d(3, 64, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1))
    (1): LeakyReLU(negative_slope=0.2, inplace=True)
    (2): Conv2d(64, 128, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1))
    (3): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (4): LeakyReLU(negative_slope=0.2, inplace=True)
    (5): Conv2d(128, 256, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1))
    (6): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (7): LeakyReLU(negative_slope=0.2, inplace=True)
    (8): Conv2d(256, 512, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1))
    (9): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (10): LeakyReLU(negative_slope=0.2, inplace=True)
    (11): Conv2d(512, 1, kernel_size=(4, 4), stride=(1, 1))
    (12): Sigmoid()
  )
)
```

Gan Outputs vs Training Inputs (500 Epochs)

Example Gan Images:



Example Training Images:



Qualitatively, pretty good! GAN images lack definition and more complex features, but match general structure well overall!

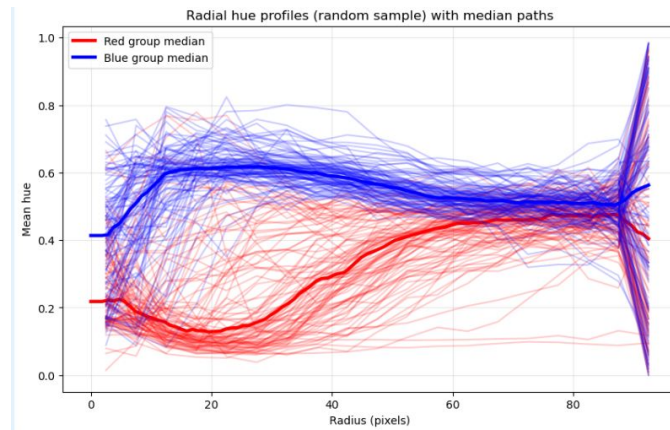
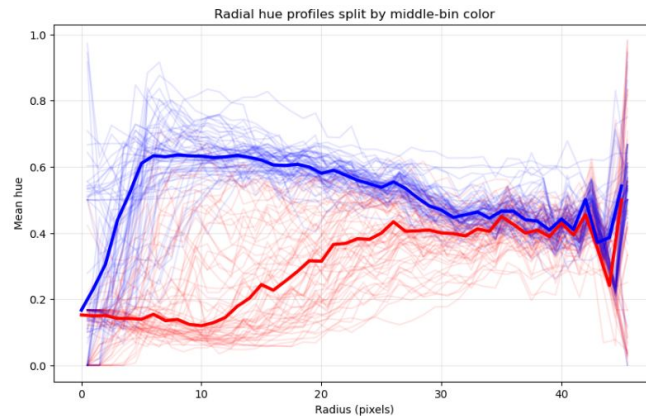
Quantitative Analysis: Radial Light Profile Binning by Color

Method:

- Separated Images into 12-px wide radial bins
- Measured average color in each bin from 0-1 (red-purple)
- Plotted color curve for each galaxy in Gan output and a random sample from training Images

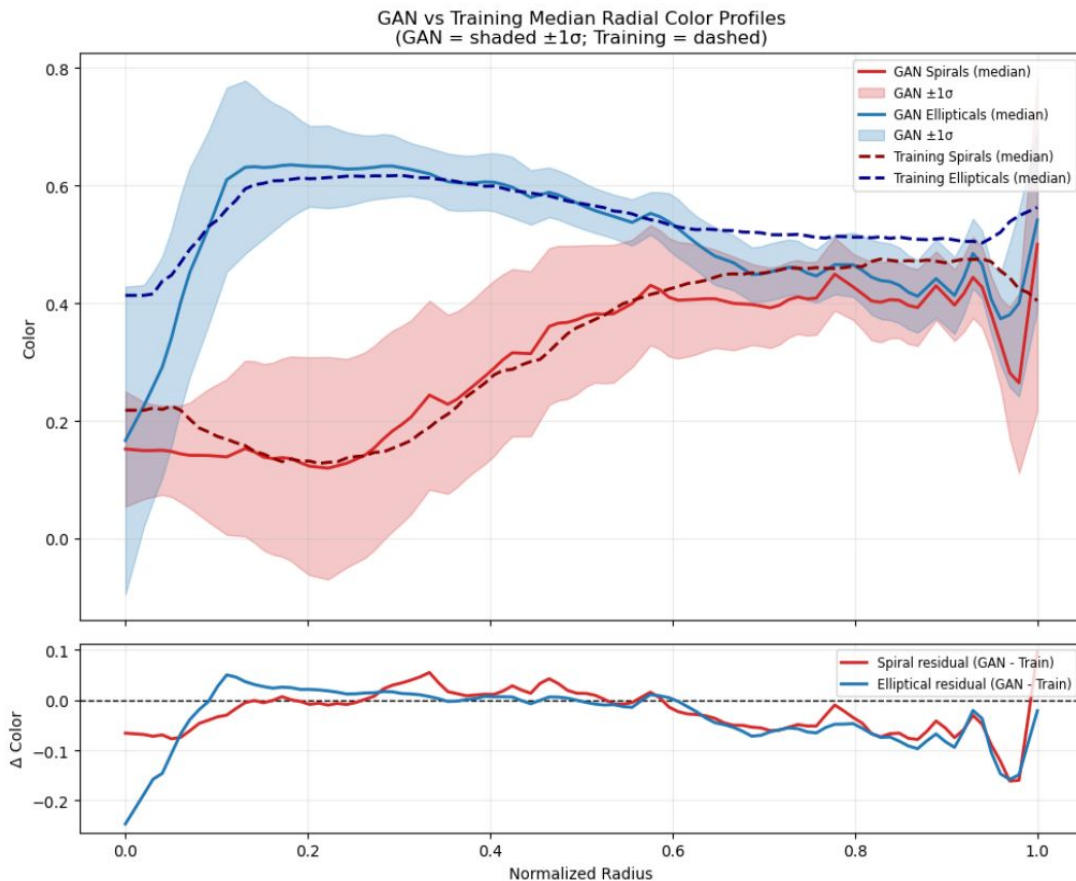
Results:

- Both samples decomposed into a bluer set and a redder set of galaxies
- Median curves matched very closely in training vs output: GAN performed well



Radial Color Binning: Error Analysis

- Qualitatively, the GAN median curves follow the training sample curves very closely
- Analyzing the residuals re-affirms our qualitative conclusions. Generally, the GAN does a good job modeling color vs radius based on the training sample
- However, we also see systematic trends at small and large radii, which correspond to high and low light intensity
- This indicated that the GAN struggles to pick up on smaller and more complex structures, especially in darker areas and in areas where the bright bulge might obscure these features.



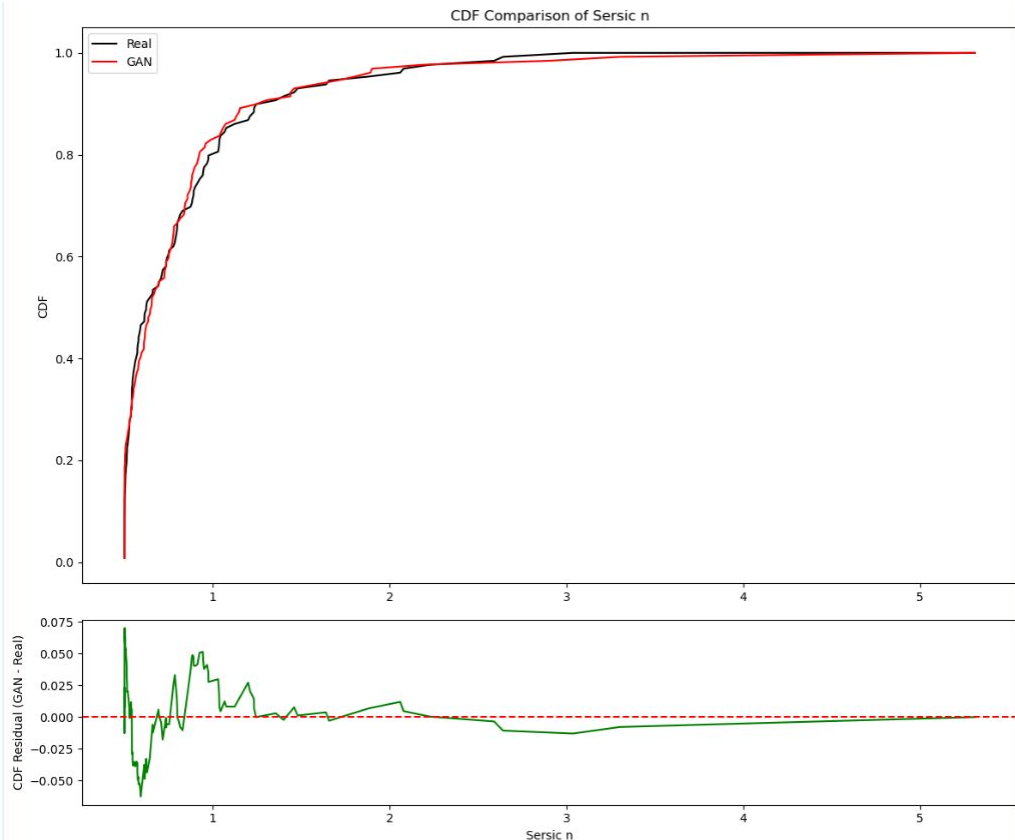
Quantitative Analysis: Sersic Profile Fitting

Method:

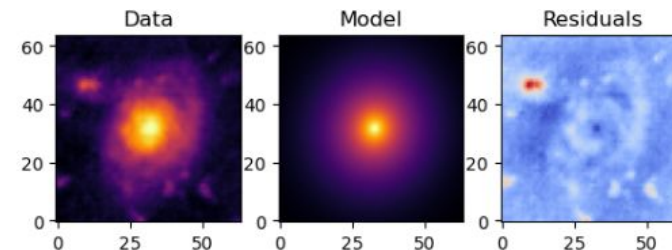
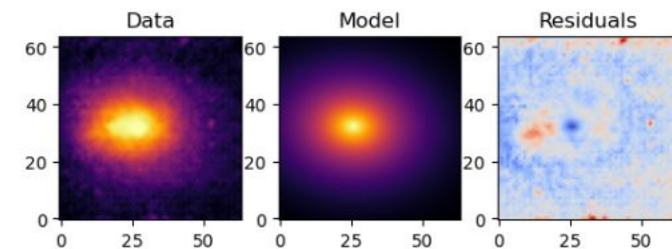
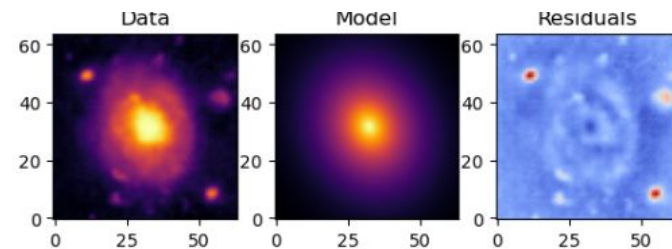
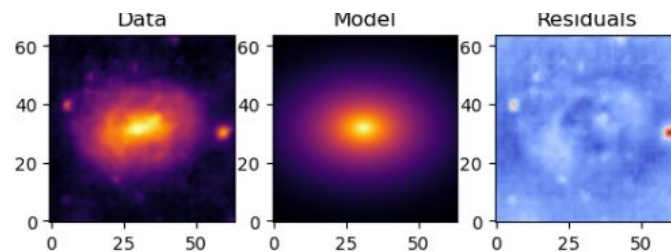
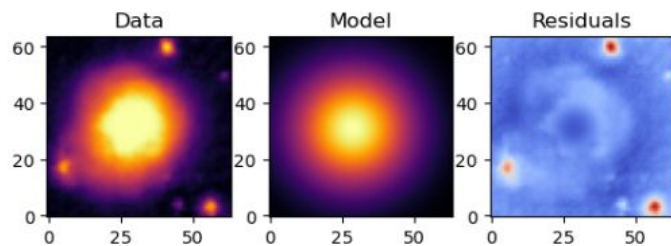
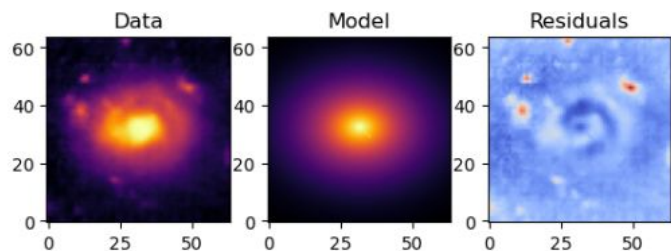
- Fit each galaxy in both sets to a two component (disk+bulge), elliptical sersic profile
- Measured and plotted scatter of sersic indices after cleaning values to reasonable ranges
- Plotted effective CDF of sersic galaxies vs training galaxies to measure agreement

Results:

- For both real and fake galaxies, we see a pattern of highest density of indices between $n = 0.5-1.0$, which decreases continuously as n increases
- Very clear agreement between CDFs, indicating again that the GAN generally did a good job emulating training galaxies, this time in terms of the PSF
- However, we see a systematic trend that the GAN has fewer galaxies with higher sersic indices, which could be another indication that it is failing to pick up faint traces of light on the outer edges of galaxies.



Sersic Model Model Fits!



Conclusion

- Qualitatively, our GAN model did a good job of reproducing real galaxies. While it failed to capture more complex features, it was able to capture several distinct morphological types and produce images that generalized galaxy light profiles very well
- Quantitatively, it performed extremely well. Both the light and color profiles of the galaxies it produced match up with the training inputs within 3σ
- Next steps include continuing to refine GAN architecture, experimenting with the applications of CNN networks to evaluate discriminator and generator accuracy, and more robust galactic analysis.

References

Meert Et. All, 2015, “A Catalogue of Two-Dimensional Photometric Decompositions in the SDSS-DR7 Spectroscopic Main Galaxy Sample: Extension to g- and i-Bands”

Radford Et. All, 2014, Unsupervised Representation Learning With Deep Convolutional General Adversarial Networks